

Venkat Chowdary Maridi

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PROFESSIONAL SUMMARY

Computer Science graduate specializing in AI Engineering and backend development, with hands-on experience building Retrieval-Augmented Generation (RAG) pipelines, LLM-powered applications, and production APIs. Built a RAG document Q&A system (FastAPI, LangChain, ChromaDB, Groq) and ContextCrunch, an open-source tool cutting LLM token usage by up to 60%. Shipped EasyAI, a prompt-validation platform with 2,000+ users in its first week. Seeking an AI Engineer role applying ML, retrieval systems, and full-stack engineering in production.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, SQL, C

AI / Machine Learning: LangChain, Retrieval-Augmented Generation (RAG), LLM Integration, Prompt Engineering, ChromaDB, Vector Embeddings, Groq API, OpenAI API, LLaMA, fastembed, Ollama, Natural Language Processing

Backend & Frameworks: FastAPI, Spring Boot, Flask, REST APIs, Node.js

Frontend: React, Next.js, Streamlit, HTML, CSS

Infrastructure & DevOps: Docker, Docker Compose, Celery, Redis, Git, GitHub, Linux

Databases: MySQL, MongoDB, ChromaDB

PROFESSIONAL EXPERIENCE

Generative AI Intern — CodeGnan IT Institute

Mar 2024 – Jul 2024

- Prototyped document Q&A and summarization tools using OpenAI and LLaMA APIs, improving contextual accuracy through iterative Retrieval-Augmented Generation (RAG) pipeline design.
- Engineered prompt structures that increased retrieval precision and contextual relevance within the RAG pipeline.
- Deployed prototypes via Streamlit, accelerating internal testing and stakeholder review cycles.
- Reduced model inference latency by 15–20% through systematic profiling and tuning of API call patterns.

PROJECTS

RAG Document Q&A System

github.com/venkatchowdari/rag-doc-qa

FastAPI • LangChain • ChromaDB • Groq (LLaMA 3.1) • fastembed • Next.js • Docker

- Architected an end-to-end Retrieval-Augmented Generation pipeline, including document ingestion, chunking, embedding generation, and grounded query response.
- Integrated Groq-hosted LLaMA 3.1 for inference and fastembed (BAAI/bge-small-en-v1.5) for embeddings, balancing latency and retrieval accuracy.
- Containerized the full stack (FastAPI backend, ChromaDB vector store, Next.js frontend) with Docker Compose for reproducible deployment.

ContextCrunch (Open Source)

github.com/venkatchowdari/contextcrunch

FastAPI • Celery • Redis • Next.js 14 • Docker • MIT License

- Built and open-sourced a self-hosted file-to-Markdown converter that reduces LLM input token consumption by 25–60% depending on file type (PDF, DOCX, XLSX, OCR images).
- Designed a 7-step Markdown optimization pipeline with automated token-savings metrics using tiktoken (cl100k_base encoding).
- Implemented an asynchronous processing queue (Celery + Redis) to handle large file conversions without blocking the API, with Flower-based task monitoring.

EasyAI

github.com/venkatchowdari/EasyAI

Next.js • OpenAI • LangChain • Vercel

- Shipped an AI-powered prompt-validation platform that classifies and screens user prompts for safety and contextual clarity before generation.
- Scaled to 2,000+ users within the first week; deployed via Vercel's serverless infrastructure for zero-downtime global access.

EDUCATION

Bachelor of Computer Science — NRI Institute of Technology

Sep 2022 – May 2026

GPA: 7.9 / 10 | Hyderabad, India

- Relevant Coursework: Data Structures, Algorithms, Machine Learning, Operating Systems, Software Design

CERTIFICATIONS

Programming Foundations (LinkedIn Learning) | Java 11 (LinkedIn Learning) | Java Data Structures (NPTEL)